目录

联合国教科文组织
信 使

社论

For a long time, local knowledge has been ignored by the scientific community and sometimes even resisted. Now, it is regaining attention, and this shift is timely: the various consequences of global mechanization, along with crises related to climate change and health issues, are prompting us to re-examine those knowledge systems, practices, and expertise that have long been marginalized.

Local knowledge is rooted in detailed observations of ecosyste ms and is passed down through generations orally, providing a valuable key to building a sustainable future. In recent years, this ty pe of knowledge has been increasingly recognized. The United Nations Declaration on the Rights of Indigenous Peoples, adopted in 2007, affirmed the rights of indigenous peoples to own, control, and develop their traditional knowledge.

UNESCO has long played a pioneering role in this field. Through the "Local and Indigenous Knowledge Systems" (LINKS) program, the organization actively promotes exchanges between indigenous knowledge holders and scientists, particularly in areas such as ecological protection, water resource management, and natural disaster prevention. Biosphere reserves are a concrete example of t his concept—bringing together researchers and local communities to jointly explore locally adapted solutions. This effort marks a si gnificant shift: science is no longer seen as a singular, top-down form of knowledge, but as a bridge for dialogue between different knowledge systems. Additionally, the Convention for the Safeguarding of the Intangible Cultural Heritage protects practices, techniques, and skills passed down through generations, recognizing their universal significance.

Local knowledge not only provides empirical data but also challenges the foundations of scientific paradigms from the perspectives of ethical values, the relationship between humans and nature, and long-term observational dimensions. In the fields of agriculture, traditional medicine, and climate science, such knowledge has long fostered numerous major innovations, whose sustainability often surpasses purely technology-driven solutions.

However, this recognition must be accompanied by safeguards—ensuring the community's free and fully informed consent, achieving equitable benefit sharing, and preventing exploitative plundering. Our goal is not merely to integrate local knowledge into the mainstream scientific system, but to promote cooperation that benefits all parties.

By promoting this type of knowledge, UNESCO reveals to us an unspoken truth: to understand and protect this world, we need to f ully integrate the knowledge and traditions of all peoples, which will bring us immeasurable benefits. True science, along with ethical standards that align with this diverse world, should provide inclusive space for the expression of all forms of human civilization, allowing the vibrant flowers of human civilization to bloom freely.

总编辑

Agnès Bardon

In the Context of Changing Times

Lagipoiva Cherelle Jackson

This indigenous climate journalist from Samoa has long reported on Pacific Islands issues for The Guardian and is currently a professor of Pacific Island Studies at Portland State University in the United States.

From the controlled burning by indigenous peoples to prevent wildfires, to the Inuit methods of weather prediction, and the 'zaï' (zaï) water harvesting techniques used in some African countries, traditional indigenous practices around the world are diverse and effective, embodying the wisdom of indigenous peoples. In the context of climate change and the ongoing loss of biodiversity, this knowledge is particularly valuable.

The embroidery work of Ecuadorian photographer Carolina Zambrano, made using Chambira palm fibers—this palm is a symbol of Amazonian indigenous craftsmanship.

Marcelo Silva de Sousa

Correspondent in Brasília, Brazil

Brazil: Insights from the Water People

The Arapaima is crucial to the survival of the Paumari people, and its population once declined sharply. In response, the Paumari in Amazonas participated in a species recovery program that combines scientific knowledge with traditional wisdom. A few years later, the Purus River has once again seen the presence of this fish species.

On the Purus River, as twilight fades, the heat still lingers over the water's surface, and an Arapaima emerges to gulp air. This deep, moist, ancient sound awakens the nearly extinct vitality of this land. " Nature is calling for our help," recalls the young fisherman Chico Paumari, "When we rushed to the lake, we found it e mpty. Our livelihood seemed to vanish before our eyes."

It's heartbreaking, I really don't know how to feed the children.

At dawn, the air is filled with a rusty, humid scent, intertwined with the smell of charcoal smoke. The Paumari people are an indigenous group of about two thousand, residing on the vast lands of Amazonas in northwestern Brazil. Their self-definition cannot be found in scientific textbooks: "We are the Water People, the only ones truly living in the water."

In 2009, the world's largest freshwater fish, the Arapaima, had only 266 adult individuals remaining, threatening the survival of the Paumari people.

For a long time, academic research and traditional knowledge have each followed their own paths without merging. Technicians use scientific methods, while indigenous communities rely on intuitive perception. A turning point emerged when both sides decided to "join forces"—this alliance was formed by Brazil's oldest indigenous peoples.

© 阿德里亚诺·甘巴里尼（Adriano Gambarini）/ 亚马孙原住民行动组织

During the fishing season, the Paumari work in shifts continuously to quickly transport the Arapaima from the lakes to the cold chain system.

When Biopiracy Breeds and Spreads

Daniel Robinson

Professor and Deputy Dean of Research at the Faculty of Arts, University of New South Wales (Australia), whose main research focus is on the regulatory mechanisms of nature and knowledge.

David Jefferson

An associate professor at the University of Canterbury (New Zealand), whose main research areas are in the intersection of biodiversity, intellectual property, and indigenous knowledge systems.

Over the past 30 years, international regulations have been committed to restricting the exploitative use of indigenous knowledge by industries to prevent such actions from harming the interests of local communities. However, despite many initiatives, cases of improper appropriation of indigenous knowledge remain common.

Globally, the ethnic groups, cultures, languages, and knowledge systems of indigenous peoples are extremely rich and diverse. These knowledge systems, developed over hundreds or even thousands of years, have made significant contributions to the crops consumed by humans today, traditional and modern medicines, as well as the fibers and materials used in home clothing. Although often referred to as "traditional knowledge," indigenous knowledge systems are actually based on empirical observation and repeated practice, and like other kno wledge systems, they continuously evolve with the assimilation of new information and changes in the local environment.

Researchers and companies often seek indigenous knowledge of the characteristics of plants, animals, and other non-human elements within ecosystems to develop ne w technologies and innovations. This process, sometimes referred to as 'bioprospecting,' is common in fields such as agriculture and food, biotechnology, pharmaceuticals, cosmetics, and forestry, leading to the development of various products sold globally.

With the help of bioprospecting, new product developers can gain numerous advantages. For example, in the pharmaceutical industry, the research and development and approval cycles are lengthy, the product development process is costly, and there are risks associated with intellectual property and regulatory protection strategies. Indigenous knowledge can help accelerate the product's entry into the market.

However, biological exploration has been criticized for "free-riding," accused of exploiting the knowledge achievements of indigenous peoples and other local groups without compensation. When indigenous knowledge is used in scientific research, product development, or commerce,

When the commercialization process fails to provide adequate recognition and the users of the knowledge do not reasonably share benefits with the holders of ancestral knowledge, such actions can be termed as " biopiracy."

The Dispute Over Benefit Sharing

Kava (scientific name: Piper methysticum) is a plant from the pepper family, native to the Pacific region. It is a typical example of the improper appropriation of indigenous peoples' and local communities' knowledge. There have been hundreds of patent applications related to kava, many of which are associated with its anti-anxiety and sedative effects, which are already well-known to the people of the Pacific region. Despite the high level of interest in kava, there are currently no agreements in place to obtain per mission or implement benefit sharing for the related research and patent claims.

From the Moroccan argan tree (scientific na me: Argania

The oil extracted from the nuts of s pinosa L. is another example. Today, argan oil is widely used in various hair and skincare products globally, and its use is essentially identical to how Amazigh (also known as Berber) women have used it for over a thousand years. Although some international giants holding relevant patents and selling argan oil products have signed benefit-sharing agreements with Amazigh women's cooperatives, other companies seem to have never shared related benefits with the Amazigh community.

Not all commercial uses of indigenous knowledge related to genetic resources constitute biopiracy; some projects achieve mutual benefits. For example, the Indjalangji-Dhidhanu tribe i n Australia collaborated with researchers from the University of Queensland to conduct research based on their indigenous knowledge about three-awned grass (a perennial tufted herb with a wide range of traditional uses). The cooperative research agreement signed by both parties included

> Nowadays, many countries require that potential users of indigenous k nowledge must first obtain the explicit consent of the knowledge holders.